Mathematical Model: Multi-Task Gaussian Process and Bayesian Matrix Completion for Distribution System State Estimation

Abstract

This document presents the mathematical formulation underlying a two-stage framework for distribution system state estimation (DSSE) under incomplete and noisy sensor measurements. In **Stage 1**, a Multi-Task Gaussian Process (MTGP) based on the Intrinsic Coregionalization Model (ICM) jointly learns the temporal dynamics across heterogeneous sensors (PMU, SCADA, AMI), providing predictive distributions that quantify epistemic uncertainty. In **Stage 2**, the MTGP predictions are assembled into a partially observed measurement matrix and completed via Bayesian low-rank matrix factorisation with physics-informed constraints derived from the linearised power-flow equations. The posterior distributions of the latent factors yield closed-form uncertainty estimates for every state variable. We detail each component's derivation, the variational update rules, and the uncertainty propagation formulae.

1 Multi-Task Gaussian Process

1.1 Problem Setting

We consider S sensor tasks. For task $s \in \{1, \dots, S\}$ we observe n_s data pairs $\{(x_i^{(s)}, y_i^{(s)})\}_{i=1}^{n_s}$, where $x_i^{(s)} \in \mathbb{R}$ denotes the (normalised) time stamp and $y_i^{(s)} \in \mathbb{R}$ denotes the (normalised) measurement value. The total number of observations is $N = \sum_{s=1}^{S} n_s$. The goal is to learn a joint probabilistic model $f_s(x)$ for all tasks simultaneously, exploiting inter-task correlations.

1.2 Intrinsic Coregionalization Model (ICM)

We model $f_s(x)$ as a zero-mean[1] multi-output GP whose covariance is given by the ICM kernel:

$$\mathrm{Cov}\big[f_s(x),\, f_t(x')\big] = B_{st}\, k_{\mathrm{time}}(x, x') + \delta_{st}\, k_{\mathrm{local},s}(x, x'), \tag{1}$$

where

- $B = LL^\top \in \mathbb{R}^{S\times S}$ is a positive semi-definite task correlation matrix parameterised through its Cholesky factor L (lower-triangular);
- $k_{\mathrm{time}}(x, x') = \sigma^2_{\mathrm{time}} \exp\!\big(-\frac{(x-x')^2}{2\ell^2_{\mathrm{time}}}\big)$ is the shared RBF kernel capturing global temporal structure;
- $k_{\mathrm{local},s}(x, x') = \sigma^2_{\mathrm{local},s} \exp\!\big(-\frac{(x-x')^2}{2\ell^2_{\mathrm{local},s}}\big)$ is a per-task RBF kernel modelling task-specific variation.

1.3 Observation Model

The noisy observation model includes per-task Gaussian noise:

$$y_i^{(s)} = f_s(x_i^{(s)}) + \varepsilon_i^{(s)}, \qquad \varepsilon_i^{(s)} \sim \mathcal{N}(0,\, \sigma^2_{\mathrm{noise},s}). \tag{2}$$

[1] In practice a shared neural-network mean function $m_\theta(x)$ is used; see Section 1.4.

Stacking all observations, the joint covariance matrix $\boldsymbol{K} \in \mathbb{R}^{N\times N}$ is block-structured. The (s,t)-block $(n_s \times n_t)$ is:

$$\boldsymbol{K}^{(s,t)} = B_{st}\,\boldsymbol{K}^{(s,t)}_{\text{time}} + \delta_{st}\big(\boldsymbol{K}^{(s,s)}_{\text{local},s} + \sigma^2_{\text{noise},s}\,\boldsymbol{I}_{n_s}\big), \tag{3}$$

where $[\boldsymbol{K}^{(s,t)}_{\text{time}}]_{ij} = k_{\text{time}}(x_i^{(s)}, x_j^{(t)})$, and similarly for the local kernel.

1.4 Shared Neural-Network Mean Function

A shared two-layer neural network $m_\theta : \mathbb{R} \to \mathbb{R}$ is employed:

$$m_\theta(x) = \boldsymbol{W}_2\,\text{ReLU}(\boldsymbol{W}_1\,x + \boldsymbol{b}_1) + b_2, \tag{4}$$

parameterised by $\theta = \{\boldsymbol{W}_1, \boldsymbol{b}_1, \boldsymbol{W}_2, b_2\}$. The same network is applied to every task, so that the residual $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{m}_\theta(\boldsymbol{x})$ is modelled by the GP.

1.5 Joint Negative Log-Marginal Likelihood

All hyperparameters $\Theta = \{\theta, L, \log\sigma_{\text{time}}, \log\ell_{\text{time}}, \{\log\sigma_{\text{local},s}, \log\ell_{\text{local},s}, \log\sigma_{\text{noise},s}\}_{s=1}^{S}\}$ are optimised by minimising the negative log-marginal likelihood (NLL):

$$\mathcal{L}(\Theta) = \frac{1}{2}\,\boldsymbol{r}^\top\boldsymbol{K}^{-1}\boldsymbol{r} + \frac{1}{2}\ln|\boldsymbol{K}| + \frac{N}{2}\ln(2\pi), \tag{5}$$

where $\boldsymbol{r} = \boldsymbol{y} - \boldsymbol{m}_\theta(\boldsymbol{x})$. Let $\boldsymbol{K} = \boldsymbol{L}_K\boldsymbol{L}_K^\top$ be the Cholesky decomposition (with a small jitter $\epsilon\boldsymbol{I}$ added for numerical stability). Then:

$$\boldsymbol{\alpha} = \boldsymbol{L}_K^{-\top}(\boldsymbol{L}_K^{-1}\boldsymbol{r}), \tag{6}$$

$$\mathcal{L} = \frac{1}{2}\,\boldsymbol{r}^\top\boldsymbol{\alpha} + \sum_{i=1}^{N}\ln[\boldsymbol{L}_K]_{ii} + \frac{N}{2}\ln(2\pi). \tag{7}$$

Gradients with respect to Θ are computed via automatic differentiation and the parameters are updated using the Adam optimiser with early stopping.

1.6 Data Normalisation

All inputs and outputs are standardised before training:

$$\tilde{x} = \frac{x - \mu_x}{\sigma_x}, \qquad \tilde{y}^{(s)} = \frac{y^{(s)} - \mu_y^{(s)}}{\sigma_y^{(s)}}, \tag{8}$$

where μ_x, σ_x are the global time mean and standard deviation, and $\mu_y^{(s)}, \sigma_y^{(s)}$ are per-task statistics. Predictions are denormalised after inference.

1.7 Predictive Distribution

Given a test point x_* for task s, define:

- Cross-covariance vector $\boldsymbol{k}_* \in \mathbb{R}^N$, where the sub-vector for task i is

$$\boldsymbol{k}_*^{(i)} = B_{is}\,\boldsymbol{k}_{\text{time}}(\boldsymbol{x}^{(i)}, x_*) + \delta_{is}\,\boldsymbol{k}_{\text{local},s}(\boldsymbol{x}^{(s)}, x_*); \tag{9}$$

- Prior test variance

$$k_{**} = B_{ss}\,k_{\text{time}}(x_*, x_*) + k_{\text{local},s}(x_*, x_*). \tag{10}$$

The predictive distribution is:

$$\mu_*^{(s)} = m_\theta(\tilde{x}_*) + \boldsymbol{k}_*^\top \boldsymbol{\alpha}, \tag{11}$$

$$\sigma_*^{(s)\,2} = k_{**} - \boldsymbol{v}^\top \boldsymbol{v}, \qquad \boldsymbol{v} = \boldsymbol{L}_K^{-1} \boldsymbol{k}_*, \tag{12}$$

and the denormalised outputs are:

$$\hat{y}_*^{(s)} = \mu_*^{(s)} \cdot \sigma_y^{(s)} + \mu_y^{(s)}, \qquad \hat{\sigma}_*^{(s)} = \sigma_*^{(s)} \cdot \sigma_y^{(s)}. \tag{13}$$

2 Physics-Constrained Variational Bayesian Matrix Completion

2.1 Problem Setting

The MTGP (or linear interpolation) predictions from Stage 1 are assembled into a measurement matrix $\boldsymbol{Z} \in \mathbb{R}^{m \times n}$, where each row corresponds to a valid (bus, phase) pair (excluding the slack bus and missing phases) and each column corresponds to a state variable:

$$\boldsymbol{Z} = \begin{bmatrix} P_1 & Q_1 & V_{r,1} & V_{i,1} & |V|_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ P_m & Q_m & V_{r,m} & V_{i,m} & |V|_m \end{bmatrix} \in \mathbb{R}^{m \times 5}. \tag{14}$$

Due to limited sensor placement, only a subset $\Omega \subseteq \{1, \ldots, m\} \times \{1, \ldots, 5\}$ of entries is directly observed. The goal is to recover the full matrix $\boldsymbol{Z}$ and quantify uncertainty for every entry.

2.2 Observation Matrix Construction

Given a three-phase distribution network with N_{bus} buses, the set of valid bus-phase combinations excludes:

- the slack bus (bus 1, all three phases);
- missing phases: buses in $\mathcal{L}_a, \mathcal{L}_b, \mathcal{L}_c$ that lack phases a, b, c respectively.

The number of valid rows is $m = 3(N_{\text{bus}} - 1) - |\mathcal{L}_a| - |\mathcal{L}_b| - |\mathcal{L}_c|$. A bijection $\phi : (\text{bus}, \text{phase}) \mapsto$ row index maps each valid bus-phase pair to a unique row.

For each sensor s providing a prediction $(\hat{z}_{ij}, \hat{\sigma}_{ij})$ from Stage 1, the observed entry $(i, j) \in \Omega$ is filled with the predicted mean $z_{ij} = \hat{z}_{ij}$, and the associated heteroscedastic noise precision is:

$$\beta_{ij}^{\text{GP}} = \frac{1}{\hat{\sigma}_{ij}^2}. \tag{15}$$

For zero-load buses, P and Q entries are set to zero with a high precision ($\beta_{ij} = 1/0.001^2$).

When using linear interpolation instead of MTGP as the Stage 1 predictor, a uniform noise precision is computed as:

$$\beta^{\text{LN}} = \frac{\text{FAD} \cdot m \cdot n}{\|\boldsymbol{Z} - \mathcal{P}_\Omega(\boldsymbol{U}_r \boldsymbol{\Sigma}_r \boldsymbol{V}_r^\top)\|_F^2}, \tag{16}$$

where $\boldsymbol{U}_r \boldsymbol{\Sigma}_r \boldsymbol{V}_r^\top$ is the rank-r truncated SVD approximation and $\mathcal{P}_\Omega$ denotes projection onto the observed set. The precision β^{LN} is assigned only to observed entries; unobserved entries receive $\beta_{ij} = 0$.

2.3 Low-Rank Generative Model

We factorise $\boldsymbol{Z} \approx \boldsymbol{A}\boldsymbol{B}^\top$, where $\boldsymbol{A} \in \mathbb{R}^{m \times d}$ and $\boldsymbol{B} \in \mathbb{R}^{n \times d}$, with latent dimension $d \ll \min(m, n)$.

Priors. Each row of $\boldsymbol{A}$ and $\boldsymbol{B}$ has an Automatic Relevance Determination (ARD) prior:

$$p(\boldsymbol{a}_i \mid \boldsymbol{\gamma}) = \mathcal{N}\big(\mathbf{0},\, \mathrm{diag}(\boldsymbol{\gamma})^{-1}\big), \qquad p(\boldsymbol{b}_j \mid \boldsymbol{\gamma}) = \mathcal{N}\big(\mathbf{0},\, \mathrm{diag}(\boldsymbol{\gamma})^{-1}\big), \tag{17}$$

where $\boldsymbol{\gamma} = (\gamma_1, \ldots, \gamma_d)^\top$ is a shared precision vector with conjugate Gamma hyper-prior $\gamma_k \sim \mathrm{Gamma}(c, d)$, where $c = d = 10^{-7}$ gives a near-uninformative prior.

Likelihood. For each observed (or augmented) entry $(i, j) \in \Omega'$:

$$p(z_{ij} \mid \boldsymbol{a}_i, \boldsymbol{b}_j, \beta_{ij}) = \mathcal{N}\big(\boldsymbol{a}_i^\top \boldsymbol{b}_j,\, \beta_{ij}^{-1}\big). \tag{18}$$

2.4 SVD-Based Initialisation

Rather than random initialisation, the latent factors are warm-started from the truncated SVD of the initial matrix (either the observation matrix $\boldsymbol{Z}$ or a full prediction matrix):

$$\boldsymbol{Z}_{\text{init}} \approx \boldsymbol{U}_d \boldsymbol{\Sigma}_d^{1/2} \cdot (\boldsymbol{\Sigma}_d^{1/2} \boldsymbol{V}_d^\top)^\top, \qquad \boldsymbol{A}^{(0)} = \boldsymbol{U}_d \boldsymbol{\Sigma}_d^{1/2}, \quad \boldsymbol{B}^{(0)} = \boldsymbol{V}_d \boldsymbol{\Sigma}_d^{1/2}, \tag{19}$$

where $\boldsymbol{U}_d, \boldsymbol{\Sigma}_d, \boldsymbol{V}_d$ are from the rank-d SVD. The initial covariances are set to $\boldsymbol{\Sigma}_{a_i}^{(0)} = \boldsymbol{\Sigma}_{b_j}^{(0)} = \alpha \boldsymbol{I}_d$ with $\alpha = 10^{-3}$.

2.5 Variational Bayesian Update Rules

The posterior of each latent vector is approximated as Gaussian. The coordinate-ascent updates are:

Update $\boldsymbol{a}_i$ (row factor):

$$\boldsymbol{\Sigma}_{a_i} = \Big(\mathrm{diag}(\boldsymbol{\gamma}) + \sum_{j:\,(i,j)\in\Omega'} \beta_{ij} \big(\boldsymbol{\mu}_{b_j} \boldsymbol{\mu}_{b_j}^\top + \boldsymbol{\Sigma}_{b_j}\big)\Big)^{-1}, \tag{20}$$

$$\boldsymbol{\mu}_{a_i} = \boldsymbol{\Sigma}_{a_i} \sum_{j:\,(i,j)\in\Omega'} \beta_{ij}\, z_{ij}\, \boldsymbol{\mu}_{b_j}. \tag{21}$$

Update $\boldsymbol{b}_j$ (column factor):

$$\boldsymbol{\Sigma}_{b_j} = \Big(\mathrm{diag}(\boldsymbol{\gamma}) + \sum_{i:\,(i,j)\in\Omega'} \beta_{ij} \big(\boldsymbol{\mu}_{a_i} \boldsymbol{\mu}_{a_i}^\top + \boldsymbol{\Sigma}_{a_i}\big)\Big)^{-1}, \tag{22}$$

$$\boldsymbol{\mu}_{b_j} = \boldsymbol{\Sigma}_{b_j} \sum_{i:\,(i,j)\in\Omega'} \beta_{ij}\, z_{ij}\, \boldsymbol{\mu}_{a_i}. \tag{23}$$

Update γ_k (ARD precision): The expected second moments under the posterior are:

$$\langle a_{ik}^2 \rangle = [\boldsymbol{\mu}_{a_i}]_k^2 + [\boldsymbol{\Sigma}_{a_i}]_{kk}, \qquad \langle b_{jk}^2 \rangle = [\boldsymbol{\mu}_{b_j}]_k^2 + [\boldsymbol{\Sigma}_{b_j}]_{kk}. \tag{24}$$

The M-step update for each ARD precision is:

$$\gamma_k = \frac{2c + m + n}{\sum_{i=1}^{m} \langle a_{ik}^2 \rangle + \sum_{j=1}^{n} \langle b_{jk}^2 \rangle + 2d}. \tag{25}$$

2.6 Linearised Power-Flow Model

The distribution network voltage is related to bus power injection through a linearised model. Let $\boldsymbol{Y}_{\text{bus}}$ be the bus admittance matrix partitioned into slack (bus 0, three phases) and non-slack nodes:

$$\boldsymbol{Y}_{\text{bus}} = \begin{bmatrix} \boldsymbol{Y}_{00} & \boldsymbol{Y}_{0n} \\ \boldsymbol{Y}_{n0} & \boldsymbol{Y}_{nn} \end{bmatrix}. \tag{26}$$

No-load voltage. With known slack-bus voltage $\boldsymbol{v}_0 = [1,\, a^2,\, a]^\top$ where $a = e^{j2\pi/3}$:

$$\boldsymbol{w} = -\boldsymbol{Y}_{nn}^{-1}\boldsymbol{Y}_{n0}\boldsymbol{v}_0 = -\boldsymbol{Z}_{nn}\boldsymbol{Y}_{n0}\boldsymbol{v}_0, \tag{27}$$

where $\boldsymbol{Z}_{nn} = \boldsymbol{Y}_{nn}^{-1}$ is the bus impedance matrix.

Complex voltage sensitivity. The linearised voltage response to power injection is $\boldsymbol{v} \approx \boldsymbol{w} + \boldsymbol{M}\boldsymbol{p}$, where:

$$\boldsymbol{M} = \left[\boldsymbol{Z}_{nn}\text{diag}(\bar{\boldsymbol{w}})^{-1} \;\middle|\; -j\,\boldsymbol{Z}_{nn}\text{diag}(\bar{\boldsymbol{w}})^{-1}\right] \in \mathbb{C}^{m\times 2m}, \tag{28}$$

and $\boldsymbol{p} = [P_1, \ldots, P_m, Q_1, \ldots, Q_m]^\top$ is the power injection vector. The factor $\text{diag}(\bar{\boldsymbol{w}})^{-1}$ arises from the linearisation of the power balance equation $S_i = V_i\bar{I}_i$ around the no-load operating point.

Voltage magnitude sensitivity. Applying a first-order expansion of $|V_i| = |w_i + \Delta V_i| \approx |w_i| + \text{Re}(\bar{w}_i\Delta V_i/|w_i|)$:

$$|\boldsymbol{v}| \approx |\boldsymbol{w}| + \boldsymbol{K}\boldsymbol{p}, \qquad \boldsymbol{K} = \text{diag}(|\boldsymbol{w}|)^{-1}\,\text{Re}\big(\text{diag}(\bar{\boldsymbol{w}})\,\boldsymbol{M}\big) \in \mathbb{R}^{m\times 2m}. \tag{29}$$

High-impedance node isolation. Floating (high-impedance) nodes whose diagonal entry $|[\boldsymbol{Z}_{nn}]_{ii}| > Z_{\text{thresh}}$ are detected and isolated by zeroing their rows and columns in $\boldsymbol{Z}_{nn}$ and resetting $w_i = 1.0$, preventing numerical instability in $\boldsymbol{M}$ and $\boldsymbol{K}$.

2.7 Physics-Constrained Iterative Algorithm

The physics model is integrated into the BMC framework through an iterative alternating scheme. Two strategies are used depending on the Stage 1 predictor.

Strategy A: Augmented observation (MTGP predictor). For buses without direct voltage measurements ($(i, j_V) \notin \Omega$ where $j_V \in \{3, 4, 5\}$ for $V_r, V_i, |V|$ columns), physics-derived pseudo-observations are injected into the augmented set $\Omega' \supseteq \Omega$ with a high precision:

$$z_{i,j_V}^{(t)} = [\boldsymbol{X}^{(t-1)}]_{i,j_V}, \quad \beta_{i,j_V} = \beta_{\text{phys}}, \tag{30}$$

where $\boldsymbol{X}^{(t-1)}$ is the reconstructed matrix from the previous iteration (with physics projection applied). This feedback loop allows the Bayesian factors to "see" the physics constraint.

Strategy B: Full physics projection (Linear predictor). All voltage columns are overwritten by the physics model after each Bayesian update:

$$\boldsymbol{v}^{(t)} = \boldsymbol{w} + \boldsymbol{M}\,\boldsymbol{p}^{(t)}, \tag{31}$$

$$|\boldsymbol{v}|^{(t)} = |\boldsymbol{w}| + \boldsymbol{K}\,\boldsymbol{p}^{(t)}, \tag{32}$$

where $\boldsymbol{p}^{(t)} = [P_1^{(t)}/(S_{\text{base}}), \ldots, Q_m^{(t)}/(S_{\text{base}})]^\top$ is extracted from the current factor product $\boldsymbol{A}^{(t)}(\boldsymbol{B}^{(t)})^\top$.

Complete iteration. Each iteration t proceeds as:

1. **Feedback:** Update pseudo-observations $z_{i,j_V}^{(t)} \leftarrow [\boldsymbol{X}^{(t-1)}]_{i,j_V}$ for $(i, j_V) \in \Omega' \setminus \Omega$;
2. **A-step:** Update each $\boldsymbol{\mu}_{a_i}, \boldsymbol{\Sigma}_{a_i}$ via Eq. (20)–(21);
3. **B-step:** Update each $\boldsymbol{\mu}_{b_j}, \boldsymbol{\Sigma}_{b_j}$ via Eq. (22)–(23);
4. **γ-step:** Update each γ_k via Eq. (25);
5. **Reconstruct:** $\boldsymbol{X}^{(t)} = \boldsymbol{A}^{(t)}(\boldsymbol{B}^{(t)})^\top$;
6. **Physics projection:** For unobserved voltage entries, overwrite with Eq. (31)–(32);
7. **Convergence check:** Stop if $\|\boldsymbol{X}^{(t)} - \boldsymbol{X}^{(t-1)}\|_F / \|\boldsymbol{X}^{(t-1)}\|_F < \varepsilon$.

2.8 Adaptive Rank Selection

The latent dimension d is selected automatically. For the MTGP predictor, two criteria are combined:

1. **Relative magnitude threshold:** retain singular values $\sigma_k > \max(\eta_{\text{noise}}, 0.02) \cdot \sigma_1$;
2. **Cumulative energy:** retain the smallest d such that $\sum_{k=1}^{d} \sigma_k^2 / \sum_k \sigma_k^2 \geq \rho$, where $\rho = 0.95$ for high noise ($\geq 10\%$) and 0.99 otherwise.

The final rank is $d = \min(d_{\text{mag}}, d_{\text{energy}})$, clamped to $[2, 4]$.

For the linear predictor, an enhanced selection combines: (i) the 95% energy criterion, (ii) the Gavish–Donoho optimal threshold $\tau = \omega(\beta) \cdot \text{median}(\sigma_k)$ where $\beta = \min(m, n)/\max(m, n)$ and $\omega(\beta) \approx 0.56\beta^3 - 0.95\beta^2 + 1.82\beta + 1.43$, and (iii) a noise-based threshold $\sigma_k > 5\eta_{\text{noise}}\sigma_1$. The final rank is $d = \max(d_{\text{energy}}, d_{\text{GD}}, d_{\text{noise}})$, clamped to $[3, 10]$.

2.9 Uncertainty Quantification

The posterior variance of the reconstructed matrix entry $\hat{z}_{ij} = \boldsymbol{\mu}_{a_i}^\top \boldsymbol{\mu}_{b_j}$ is obtained in closed form.

Since $\boldsymbol{a}_i \sim \mathcal{N}(\boldsymbol{\mu}_{a_i}, \boldsymbol{\Sigma}_{a_i})$ and $\boldsymbol{b}_j \sim \mathcal{N}(\boldsymbol{\mu}_{b_j}, \boldsymbol{\Sigma}_{b_j})$ are assumed independent a posteriori, we have:

$$\mathbb{E}[\hat{z}_{ij}^2] = \mathbb{E}[\boldsymbol{a}_i^\top \boldsymbol{b}_j \boldsymbol{b}_j^\top \boldsymbol{a}_i] = \text{tr}\big((\boldsymbol{\mu}_{a_i}\boldsymbol{\mu}_{a_i}^\top + \boldsymbol{\Sigma}_{a_i})(\boldsymbol{\mu}_{b_j}\boldsymbol{\mu}_{b_j}^\top + \boldsymbol{\Sigma}_{b_j})\big), \tag{33}$$

$$(\mathbb{E}[\hat{z}_{ij}])^2 = (\boldsymbol{\mu}_{a_i}^\top \boldsymbol{\mu}_{b_j})^2. \tag{34}$$

Expanding and cancelling the $(\boldsymbol{\mu}_{a_i}^\top \boldsymbol{\mu}_{b_j})^2$ term yields:

$$\boxed{\text{Var}(\hat{z}_{ij}) = \underbrace{\text{tr}\big(\boldsymbol{\Sigma}_{a_i}\boldsymbol{\Sigma}_{b_j}\big)}_{\text{interaction noise}} + \underbrace{\boldsymbol{\mu}_{a_i}^\top \boldsymbol{\Sigma}_{b_j} \boldsymbol{\mu}_{a_i}}_{\text{B-uncertainty projected onto A-mean}} + \underbrace{\boldsymbol{\mu}_{b_j}^\top \boldsymbol{\Sigma}_{a_i} \boldsymbol{\mu}_{b_j}}_{\text{A-uncertainty projected onto B-mean}}}. \tag{35}$$

3 Posterior Structural Probability Inference

3.1 Prediction Interval Construction

Under the Gaussian posterior assumption, the $(1 - \alpha)$ prediction interval for state variable $\hat{z}_{ij}$ is:

$$\hat{z}_{ij} \pm z_{1-\alpha/2} \sqrt{\text{Var}(\hat{z}_{ij})}, \tag{36}$$

where $z_{1-\alpha/2}$ is the standard normal quantile (e.g., $z_{0.975} = 1.96$ for 95% confidence).

Prediction Interval Coverage Probability (PICP):

$$\text{PICP} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\big[z_i^{\text{true}} \in [\hat{z}_i - z_{1-\alpha/2}\hat{\sigma}_i,\ \hat{z}_i + z_{1-\alpha/2}\hat{\sigma}_i]\big]. \tag{37}$$

A well-calibrated model achieves PICP $\approx (1 - \alpha) \times 100\%$.

Mean Prediction Interval Width (MPIW):

$$\text{MPIW} = \frac{1}{n} \sum_{i=1}^{n} 2\, z_{1-\alpha/2}\, \hat{\sigma}_i. \tag{38}$$

A smaller MPIW indicates tighter (more informative) intervals.

3.2 Voltage Angle Uncertainty Propagation

The voltage angle $\theta = \arctan(V_i/V_r)$ is a nonlinear function of the estimated real and imaginary voltage components. Its uncertainty is obtained via the delta method (first-order Taylor expansion):

$$\frac{\partial \theta}{\partial V_r} = \frac{-V_i}{V_r^2 + V_i^2}, \qquad \frac{\partial \theta}{\partial V_i} = \frac{V_r}{V_r^2 + V_i^2}. \tag{39}$$

Assuming V_r and V_i are uncorrelated a posteriori (valid since they occupy different columns of the factorised matrix):

$$\sigma_\theta \approx \frac{\sqrt{V_r^2\, \sigma_{V_i}^2 + V_i^2\, \sigma_{V_r}^2}}{V_r^2 + V_i^2}, \tag{40}$$

where $\sigma_{V_r}, \sigma_{V_i}$ are the posterior standard deviations from Eq. (35).

3.3 Point-Estimate Error Metrics

The following metrics are used to evaluate reconstruction quality:

Mean Integrated Absolute Error (MIAE): For power and voltage angle:

$$\text{MIAE} = \frac{1}{n} \sum_{i=1}^{n} |z_i^{\text{true}} - \hat{z}_i| \times 100\%. \tag{41}$$

Mean Absolute Percentage Error (MAPE): For voltage magnitude:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^{n} \frac{|z_i^{\text{true}} - \hat{z}_i|}{|z_i^{\text{true}}|} \times 100\%. \tag{42}$$